

Question 12

Problem Statement

A decimation program produces **28 output samples** from an input of **96 samples** with decimation factor **4**. Identify the coding error.

Step 1: Given Data

- Input Samples (N) = 96
- Decimation Factor (M) = 4
- Reported Output Samples = 28

Step 2: Formula

Output Samples = Input Samples / Decimation Factor

Step 3: Correct Calculation

Output Samples = $96 / 4 = 24$ **samples**.

Correct Output = 24 samples

Step 4: Error Identification

The program produced **28 samples** instead of **24 samples**. Hence, the decimation process has been implemented incorrectly.

Root Cause Analysis

1. The decimation factor ($M = 4$) was not applied correctly in the program.
2. The sample indexing skipped the wrong number of samples.
3. Extra samples were included due to an incorrect loop or array index.
4. The output length calculation was coded incorrectly.

Correction

Select every **4th** input sample and discard the remaining samples. For 96 input samples with $M = 4$, the output must contain exactly **24 samples**.

Conclusion

The correct number of output samples after decimation is **24**, not **28**. Correct indexing and proper application of the decimation factor ensure accurate sampling rate reduction in DSP.